

Project Portfolio

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Introduction

My name is Vatsal Gohel. I currently serve as Director of Institutional Research at Clinton Community College, where I lead data strategy, accreditation, and institutional planning. I hold a Master of Science in Data Analytics (STEM) from Clark University and bring extensive experience extracting actionable insights from complex datasets to drive strategic decision-making and optimize organizational processes.

Professional Summary

I am a results-driven institutional research professional with a strong technical analytics background and the ability to translate complex data into clear, evidence-based insights for executive leadership and external stakeholders. In my current role at Clinton Community College, I lead all institutional research, strategic planning, and academic assessment functions, serve as Accreditation Liaison Officer (ALO) for MSCHE, and oversee accreditation reporting and compliance. I represent the college at SUNY and professional IR organizations and conferences.

Prior to this, I served as a Data Analyst at Clinton Community College — leading all NYSED, SUNY, IPEDS, and MSCHE data submissions and developing a KPI tracking dashboard for the college's Strategic Planning initiative. I have additionally worked as a Data Analyst at Pinnacle Software Solutions and DynaTech Systems Inc., building scalable Power BI dashboards and SQL-based reporting solutions. At Clark University I served as Teaching Assistant supporting 60+ graduate students. At Numerator, I progressed from Data Quality Associate to Research Analyst and was recognized as Employee of the Year for high-impact delivery across global CPG client portfolios.

Tools Used and Known

I am proficient in a wide range of tools spanning analytics, visualization, and institutional research:

- Visualization & BI: Power BI, Tableau — interactive dashboards, KPI tracking, institutional reporting
- Programming: Python (Pandas, NumPy, Seaborn, Plotly, Matplotlib), R — statistical analysis, machine learning, ETL scripting
- Data Management: SQL / SQL Server, Advanced Excel, ETL processes, data warehousing
- Institutional Research: IPEDS, NYSED, SUNY, MSCHE reporting; accreditation compliance; program review; assessment
- Cloud & Infrastructure: Azure, AWS; SSRS (SQL Server Reporting Services)
- Certifications: Google Analytics, Power BI for Data Analysts (Microsoft), BCG Data Science Job Simulation, AWS Cloud Practitioner CLF-C02, Anaconda Python for Data Science, SQL for Data Analysis, Excel: Advanced Formulas & Functions

10. E-Commerce Sales and Customer Analysis

Project Title

Unveiling Consumer Insights and Regional Dynamics in E-Commerce

Problem Statement

The e-commerce industry generates vast transactional data, yet many platforms fail to systematically mine it for strategy. Without clear understanding of which customer segments drive revenue, how discounts affect margins, and which regions represent growth opportunities, expansion and pricing decisions remain guesswork.

Project Description

Objective: To analyse customer behaviour, discount impact, and regional sales performance to identify growth opportunities and provide actionable pricing and expansion recommendations.

Scope:

- Analyse customer orders, payment methods, demographics, and regional sales across categories.
- Evaluate the financial impact of discount strategies on revenue and margin.
- Identify high-performing and under-performing regions and product categories.

Deliverables:

- Reports on customer preferences and sales trends by region and category.
- Discount impact analysis and pricing strategy recommendations.
- Insights supporting potential offline expansion strategy.

Methodology

Tools & Technologies: Python (Pandas, NumPy, Matplotlib, Seaborn), Excel, Jupyter Notebook, Tableau

Process:

- Data Cleaning: handled missing values; corrected inconsistent status labels; removed duplicates; ensured integrity.
- EDA: visualised sales trends by region, category, and time; analysed payment method distribution and customer demographics.
- Discount Analysis: calculated average discount levels by category; modelled revenue trade-off of discount depth vs. volume uplift.
- Regional Analysis: mapped geographic performance; identified high-growth and lagging regions.
- Reporting: compiled findings into stakeholder-ready visualisations and strategic recommendations.

Role and Responsibilities

Role: Team Data Analyst

- Led EDA and statistical analysis to derive regional and customer-segment insights.
- Compiled and presented findings in the stakeholder report.

Challenges and Solutions

- Ambiguous data fields (inconsistent order and billing status labels) — thorough validation and label standardisation during cleaning.
- Privacy compliance for customer-level data — anonymisation techniques applied to sensitive fields before analysis.

Results and Impact

- Significant regional performance variation identified: specific markets over- or under-indexed relative to population base.
- Deep discounts (>30%) found to reduce net revenue in most categories despite volume increases — prompted a recommended discount cap policy.
- Traditional payment methods dominated — flagged as a checkout friction point for optimisation.

Conclusion and Learnings

Key learning: discount analysis requires margin modelling, not just revenue modelling — top-line volume uplift can mask bottom-line deterioration. Regional segmentation consistently surfaces non-obvious growth opportunities even at a coarse level.

Link: [E-Commerce Sales and Customer Analysis](#)